

UltraUPConvNet: A UPerNet- and ConvNeXt-Based Multi-Task Network for Ultrasound Tissue Segmentation and Disease Prediction

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Abstract. Ultrasound imaging is widely used in clinical practice due to its cost-effectiveness, mobility, and safety. However, current AI research often treats disease prediction and tissue segmentation as two separate tasks and their model requires substantial computational overhead. In such a situation, we introduce UltraUPConvNet, a computationally efficient universal framework designed for both ultrasound image classification and segmentation. Trained on a large-scale dataset containing more than 9,700 annotations across seven different anatomical regions, our model achieves state-of-the-art performance on certain datasets with lower computational overhead. Our model weights and codes are available at <https://github.com/yyx1123/UltraUPConvNet>.

Keywords: Promptable Learning · Ultrasound · Universal Model.

1 Introduction

Ultrasound imaging is widely adopted in clinical practice due to its cost-effectiveness, mobility, and safety. As deep learning models have shown strong performance in applications such as computer vision and natural language processing, deep learning models for ultrasound have also shown great potential. Recently, General Medical Artificial Intelligence (GMAI) models [1] have demonstrated robust generalization capabilities across a wide range of tasks and datasets. Although some GMAI models have been proposed, they typically require substantial computational overhead and have relatively complex architectures, highlighting the need for a model specifically for ultrasound with lower computational overhead and simpler architectures.

Recent advances in deep learning models for ultrasound imaging have addressed some specific tasks, such as breast tumor prediction [2]. However, when faced with new tasks or datasets, they overlook intrinsic dataset task relationships and need extensive retraining. The focus is now on universal models, known for their versatility in segmentation tasks across domains. At the same time, some universal models like MedSAM [3], SAM-Med2D [4], UniUSNet [5] and SAMUS

[6] are characterized by the heavy stacking of Transformer blocks, leading to substantial computational costs and increased architectural complexity.

Most methods perform well in segmentation while struggling with classification. But our ConvNeXt-UperNet-based network can do classification task more effectively. For ultrasound’s complexity, we propose fully automated prompts that encode prior knowledge.

Therefore, we propose UltraUPConvNet, a universal and promptable framework for ultrasound that addresses multiple clinical tasks. Our contributions include:

- Versatile Model Framework: Our framework uses four model prompts to obtain better performance and flexibility in various clinical tasks.
- Efficient and Simple Model Architecture: Our framework does not adopt the currently popular Transformer architecture, but instead relies entirely on convolutional operations, resulting in lower computational complexity and a simpler model structure.
- Extensive Experiments: The model outperforms single dataset and ablated versions, achieving state-of-the-art performance and strong generalization to new domains.

2 Method

2.1 Multi-task Paradigm

Multi-task learning is a machine learning method that enhances the generalization ability of each task by simultaneously learning multiple related tasks within the same model. In our specific task, the proposed framework is required to perform both disease prediction and tissue segmentation, which correspond to a classification task and a segmentation task, respectively.

Therefore, the proposed model incorporates two dedicated decoders, one for classification and the other for segmentation. The overall loss is computed in an alternating fashion, where segmentation and classification batches are processed separately during each training epoch. This strategy avoids task interference and maintains task-specific supervision, while still benefiting from a shared feature extractor.

2.2 Model Architectures

UltraUPConvNet (Fig. 1) is a general encoder-decoder model. It also uses prompts to simultaneously help with multiple ultrasound tasks like segmentation and classification. The encoder extracts features, while task-specific decoders are enhanced by four types of prompts—nature, position, task, and type—added to extracted features via prompt projection embedding, boosting the model’s versatility and performance.

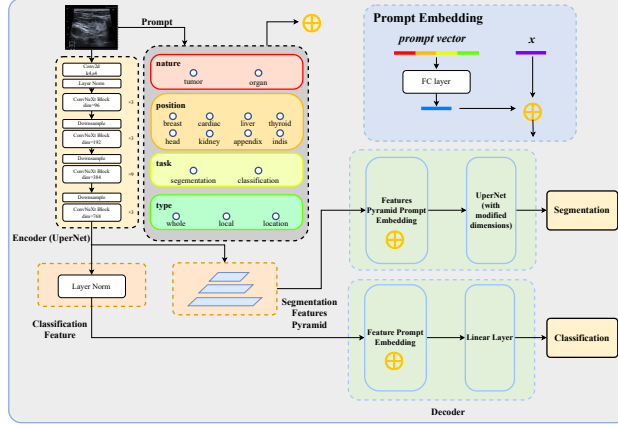


Fig. 1. The architecture of the proposed UltraUPConvNet.

Encoder Inspired by UniUSNet’s good performance, we would like to try a better and simpler backbone without using Transformer. We finally chose ConvNeXt [7]. ConvNeXt is an efficient convolutional neural network (CNN) model whose design integrates the advantages of traditional convolutional neural networks and Transformer models, aiming to provide more efficient computational performance while maintaining high accuracy. To make our model lightweight, we chose ConvNeXt-Tiny, which is the most compact variant in the ConvNeXt family while still maintaining competitive performance.

Prompting Strategy In our specific task, four prompts—nature, position, task, and type—are used to provide information about images and tasks. This method can possibly enhance the model’s flexibility and interpretability. Learning from Vision Transformers (ViT) [8] and [5], our work defines four prompt one-hot vectors and project them into specific dimensions using fully connected (FC) layers in both classification and segmentation tasks.

Decoder Since we want to have a lightweight model, we choose UperNet [9] as our segmentation decoder. UperNet is built upon the Feature Pyramid Network (FPN) and PPM (Pyramid Pooling Module). It is proven to excel at semantic segmentation of medical images [10]. Specifically, FPN combines high-resolution low-level features with semantically rich high-level features through a top-down pathway and lateral connections. To match the input dimensionality, we also modified the dimensions of UperNet accordingly.

2.3 Loss Function

Classification For the segmentation task, we adopt a compound loss that balances between pixel-wise classification and region-wise consistency. Specifically,

we use a weighted combination of Cross Entropy Loss and Dice Loss. Given the model’s output x_{seg} and the ground truth label y , the total segmentation loss is defined as:

$$\mathcal{L}_{\text{seg}} = 0.4 \cdot \mathcal{L}_{\text{CE}}(x_{\text{seg}}, y) + 0.6 \cdot \mathcal{L}_{\text{Dice}}(x_{\text{seg}}, y)$$

Here, $\mathcal{L}_{\text{CE}}$ is the standard cross-entropy loss, while $\mathcal{L}_{\text{Dice}}$ is the soft Dice loss computed over the softmax outputs of the segmentation map. This combination enhances both per-pixel accuracy and overall region segmentation quality.

Segmentation The classification task is designed to handle both binary (2-way) and multi-class (4-way) scenarios within a single training loop. During each iteration, the number of classes required for each sample is indicated by a batch-level metadata flag. Accordingly, two separate classifier heads are employed: one for 2-way classification and another for 4-way classification. The total classification loss is computed as:

$$\mathcal{L}_{\text{cls}} = \begin{cases} \mathcal{L}_{\text{CE-2}}(x_{\text{cls-2}}, y) & \text{if 2-way} \\ \mathcal{L}_{\text{CE-4}}(x_{\text{cls-4}}, y) & \text{if 4-way} \end{cases}$$

where $x_{\text{cls-2}}$ and $x_{\text{cls-4}}$ denote the classification outputs from the respective classifier heads. If a batch contains both types of tasks, the final loss is the sum of the two. This design allows for flexible handling of mixed-task datasets.

Overall Loss Strategy The overall loss is computed in an alternating fashion, where segmentation and classification batches are processed separately within each training epoch. Specifically, segmentation loss and classification loss are computed from their respective data loaders in separate loops. This strategy enables task-specific optimization while allowing the model to learn shared representations across tasks.

To balance the gradient contributions from both tasks, the classification loss is scaled by a weighting coefficient λ_{cls} during backpropagation. The final loss for backpropagation is defined as:

$$\mathcal{L}_{\text{final}} = \begin{cases} \lambda_{\text{seg}} \cdot \mathcal{L}_{\text{seg}}, & \text{for segmentation batches} \\ \mathcal{L}_{\text{cls}}, & \text{for classification batches} \end{cases}$$

This weighted design helps stabilize multi-task training by preventing the classification task from dominating or under-contributing to the optimization process. In our experiments, λ_{cls} is set to 10, empirically based on validation performance.

3 Experimental Results

3.1 Implementation Details

The dataset, as shown in Table. 1, was split into training, validation, and testing sets (7:1:2). We also used ConvNeXt’s official pre-trained weights. Learning from

[5], we trained our models for 200 epochs using AdamW (initial learning rate: $2e-5$) with data augmentation techniques like random flipping, rotation (-20° to 20°), and cropping. Since our model is highly lightweight, it can be trained using an RTX 2060 with only 6GB of VRAM. Implementation was done in PyTorch.

Table 1. DATASET DESCRIPTION.

Dataset	Public?	Position	Image Num	Annotation
BUSI [11]	✓	Breast	780	Classification, Segmentation
BUSIS [12]	✓	Breast	562	Segmentation
BUS-BRA [13]	✓	Breast	1875	Classification, Segmentation
Fatty-Liver [14]	✓	Liver	550	Classification
kidneyUS [15]	✓	Kidney	534	Segmentation
DDTI [16]	✓	Thyroid	466	Segmentation
Fetal HC [17]	✓	Head	999	Segmentation
CAMUS [18]	✓	Cardiac	500	Segmentation
Appendix [19]	✓	Appendix	474	Classification
Appendix	✗	Appendix	67	Classification
Breast	✗	Breast	150	Classification, Segmentation
Breast-luminal	✗	Breast	236	Classification, Segmentation
Cardiac	✗	Cardiac	76	Segmentation
Fetal Head	✗	Head	121	Segmentation
Kidney	✗	Kidney	67	Segmentation
Liver	✗	Liver	103	Classification
Thyroid	✗	Thyroid	428	Segmentation

3.2 Results

Comparison with SOTA We compared our model with state-of-the-art (SOTA) models, like SAMUS and UniUSNet. SAMUS is a SAM variant for ultrasound data and UniUSNet is based on a modified Swin-Unet. From [5], we get SAMUS’s and UniUSNet’s results trained on BroadUS-9.7K Dataset. From Table. 2, SAMUS’s segmentation performance (80.01%) is lower than UniUSNet (85.80%), while SAMUS has better segmentation performance on more datasets. Compared to UniUSNet, our automatic prompt model, with 29.9% fewer parameters, achieves better both segmentation (90.28%) and classification (89.95%) results.

Ablation Study We also trained our UltraUPConvNet model without prompts to assess whether prompts can enhance performance. The segmentation results in Fig. 2 reveal that UltraUPConvNet performs well on the segmentation task. From Table. 2, ablation study reveals that UltraUPConvNet (90.11%) outperforms the ablation version (89.90%), proving the effectiveness of prompts.

Table 2. OVERALL PERFORMANCE COMPARISON ON SEVERAL DATASETS.

Dataset	SAMUS (Interactive)	UniUSNet (Automatic)	UltraUPConvNet w/o prompt (Automatic)	UltraUPConvNet (Automatic)
Params	130.10M	86.29M	60.44M	60.48M
BUS-BRA	83.88%	75.59%	88.37%	88.46%
BUSIS	88.89%	87.84%	92.23%	91.33%
CAMUS	72.44%	92.58%	94.62%	94.71%
DDTI	69.70%	66.06%	79.25%	80.55%
Fetal_HC	97.60%	96.69%	97.72%	97.11%
KidneyUS	67.54%	96.03%	90.02%	89.49%
Seg Average	80.01%	85.80%	90.37%	90.28%
Appendix	/	52.84%	75.89%	77.30%
BUS-BRA	/	80.39%	90.96%	92.02%
Fatty-Liver	/	89.36%	100%	100.00%
Cls Average	/	74.20%	88.95%	89.77%
Total Average	/	81.93%	89.90%	90.11%

**Fig. 2.** Some examples of segmentation result. From left to right: original image, UltraUPConvNet w/o prompt, Prompt and ground truth.

4 Discussion and Conclusion

We propose a lightweight universal model for medical ultrasound imaging that integrates four prompts. Experiments on some public datasets validate its effectiveness. We will further assess the model’s adaptability in our future work.

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